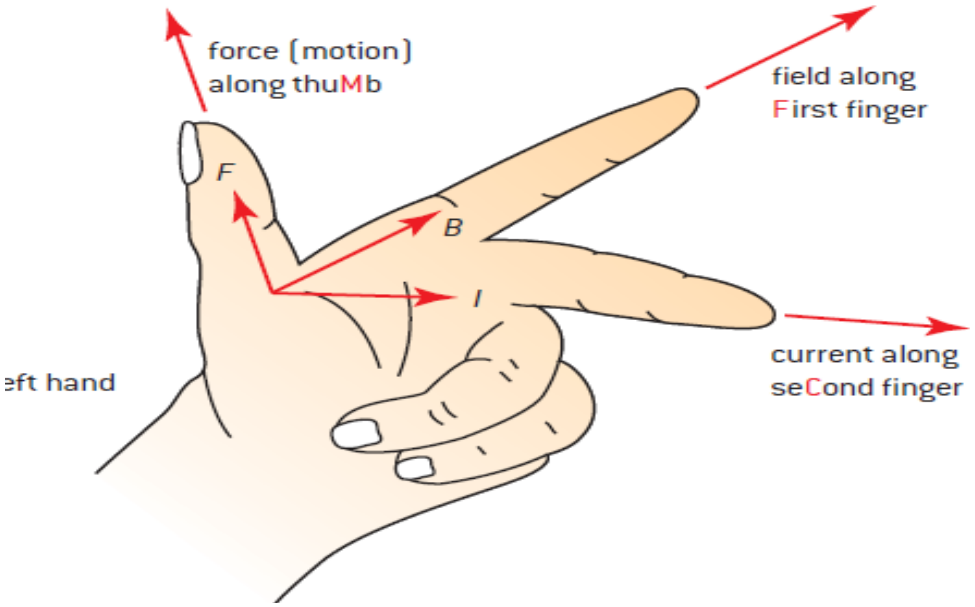
Lesson 2

Magnetic Force

Lesson Objective: You will be able to:

- Determine the direction of magnetic force, current, field
- Describe and differentiate motor, generator, transformer
- Calculate number of turns, voltage current in transformer.
- Experiment: Make simple motor.

1- Force on Current carrying Conductor in Magnetic Field



Whenever a current carrying conductor comes under a magnetic field, there will be a force acting on the conductor. It is found that whenever a current carrying conductor is placed inside a magnetic field, a force acts on the conductor, in a direction, perpendicular both to the direction of the electric current and the magnetic field. The direction of magnetic force is given using the **Fleming Left Hand rule**.

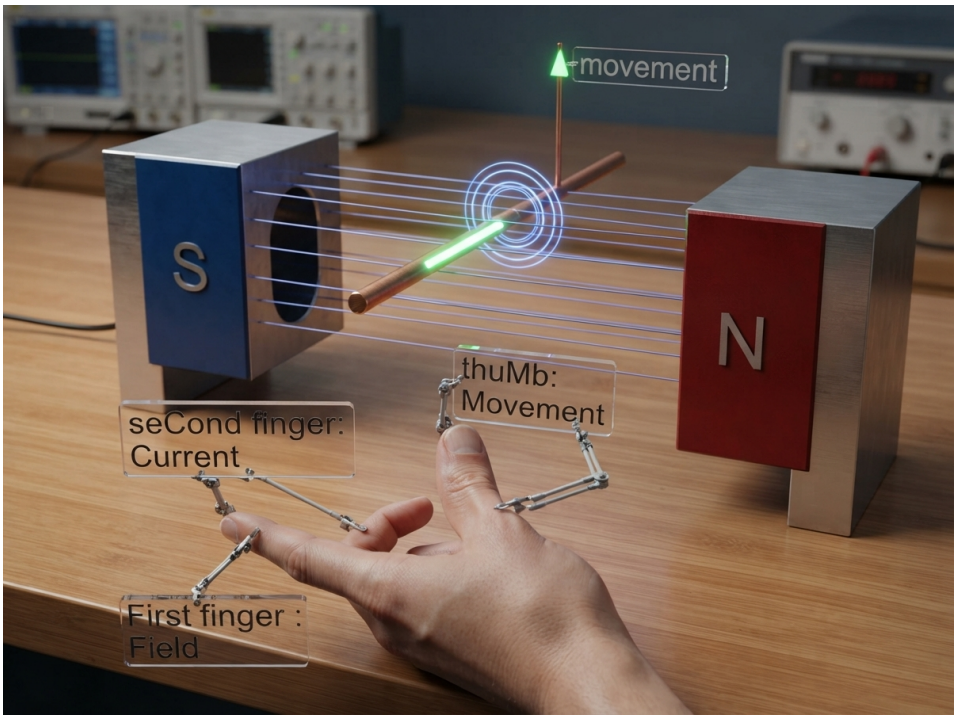
Fleming Left Hand rule is mainly applicable for electric motor.

Outstretch the three fingers of the left hand perpendicular to each other.

+Point your first finger in the direction of magnetic field

+Second finger in the direction of current

+Thumb gives the direction of force.



Experiment (with current-carrying conductors placed at right angle to magnetic field) shows that the force acting depends on:

-The strength of the magnetic field.

-The current.

-The length of conductor.

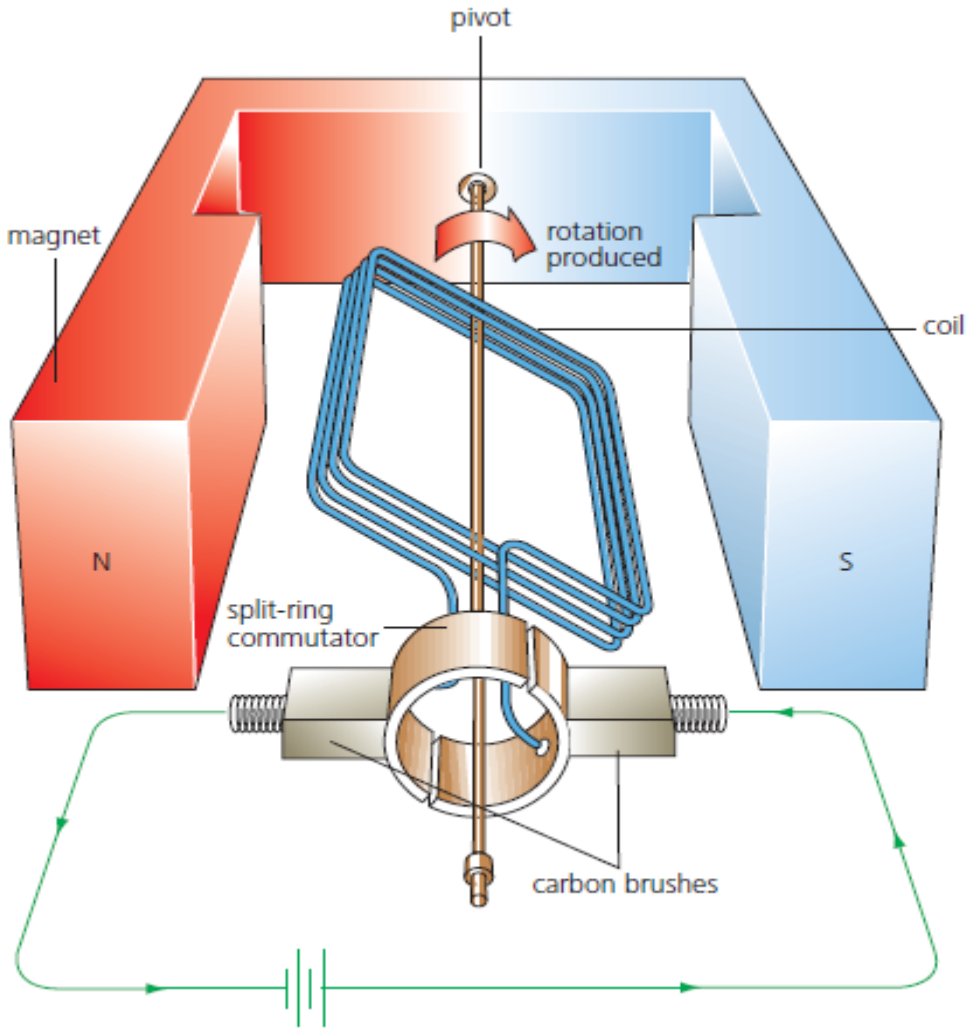
2-Principle of DC Motor

The electric motor is a device which converts electrical energy to mechanical energy. Every DC motor has six basic parts: axle, rotor (armature), stator, commutator, field magnet(s), and brushes.

In order to understand the **operating principle of dc motor** we need to first look into its constructional feature.

The very basic construction of a dc motor contains a current carrying armature which is connected to the supply end through commutator segments and brushes and placed within the north south poles of a permanent or an electro-magnet as shown in the diagram.

Now to go into the details of the **operating Principle of dc motor** it is important that we have a clear understanding of Fleming's left hand rule to determine the direction of force acting on the armature conductors of dc motor.



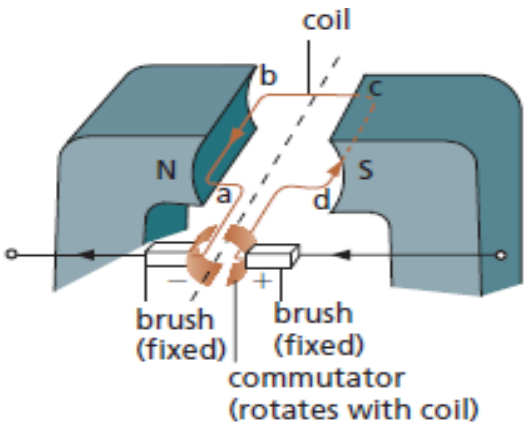
Fleming Left Hand rule is mainly applicable for electric motor.

Outstretch the three fingers of the left hand namely the first finger, middle finger, and the thumb such that they are mutually perpendicular to each other. Now point

☞ *your first finger in the direction of magnetic field*

☞ *middle finger in the direction of current then the thumb gives the direction of force*.

We find that side **ab** experiences an upward force and side **cd** a downward force.
(No forces act on ad and **bc** since they are parallel to the field.) These two forces form a couple (forces in pair) which rotates the coil in a clockwise direction until it is vertical.



The brushes are then in line with the gaps in the commutator and the current stops. However, because of its inertia, the coil overshoots the vertical and the commutator halves change contact from one brush to the other. This reverses

the current through the coil and so also the directions of the forces on its sides. Side ab is on the right now, acted on by a downward force, while cd is on the left with an upward force. The coil thus carries on rotating clockwise.

The more turns there are on the coil, or the larger the current through it, the greater is the couple on the coil and the faster it turns. The coil will also

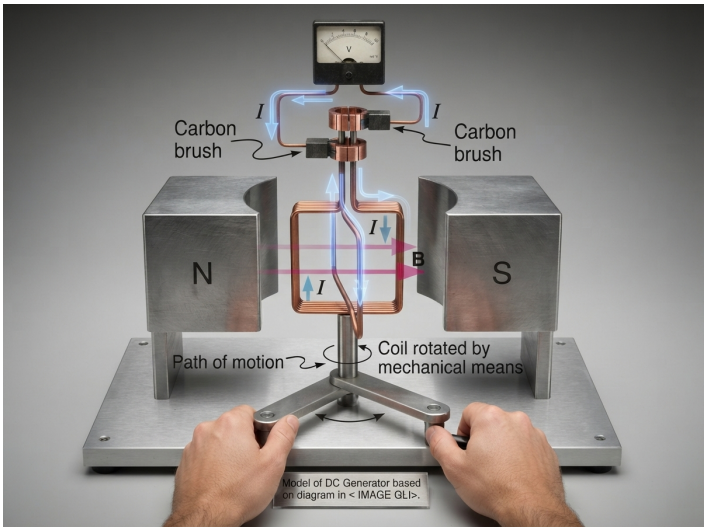
turn faster if the strength of the magnetic field is increased.

The uses of DC motor: the windscreen wipers, toy car, …….

3- Simple a.c. generator(alternator)

The effect of producing electricity from magnetism is called electromagnetic induction and was discovered in 1831 by Faraday. Faraday found, and it can be shown with apparatus like that in Figure below, that the induced voltage increases with increase of

1. The speed of motion of the magnet or coil
2. The number of turns on the coil
3. The strength of the magnet.

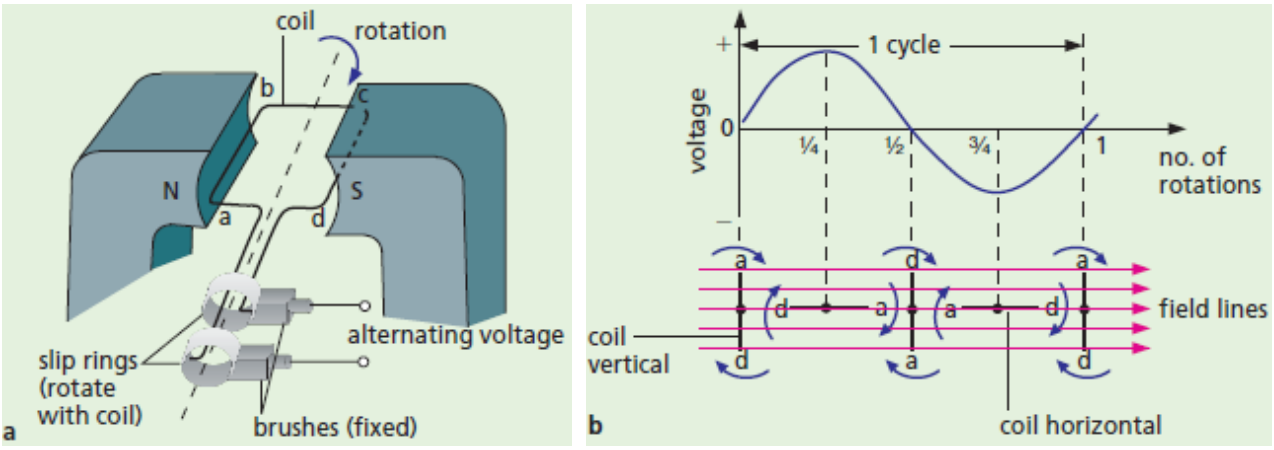


In electricity generation, an **electric generator** is a device that converts mechanical energy to electrical energy.

The simplest alternating current (a.c.) generator consists of a rectangular coil between the poles of a magnet. The ends of the coil are joined to two slip rings on the axle and against which carbon brushes press.

When the coil is rotated it cuts the field lines and a voltage is induced in it. Figure below shows how the voltage varies over one complete rotation.

As the coil moves through the vertical position with **ab** uppermost, ab and **cd** are moving along the lines (bc and da do so always) and no cutting occurs. The induced voltage is zero.



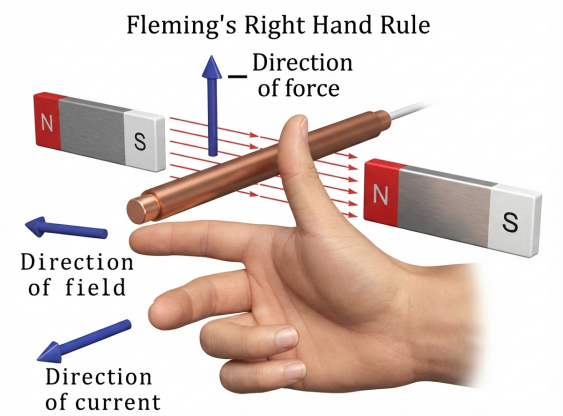
Faraday's law of electromagnetic induction, whenever a conductor moves inside a magnetic field, there will be an induced current in it. If this conductor is forcefully moved inside the magnetic field, there will be a relation between the direction of applied force, magnetic field and the induced current. This relation among these three directions, is determined by **Fleming Right Hand Rule.**

+First finger represents the direction of the field.

+Thumb points in the direction of motion or applied force.

applied force.

+Second finger points in the direction of the induced current.

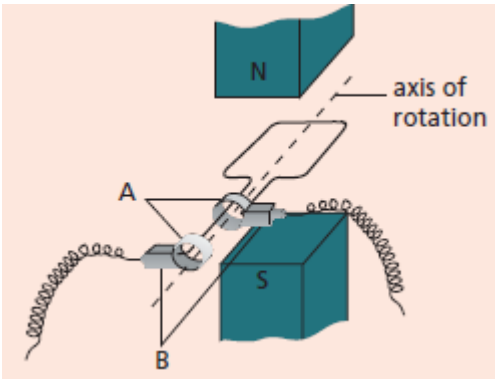


Question

1-A simple generator is shown in Figure 43.11.

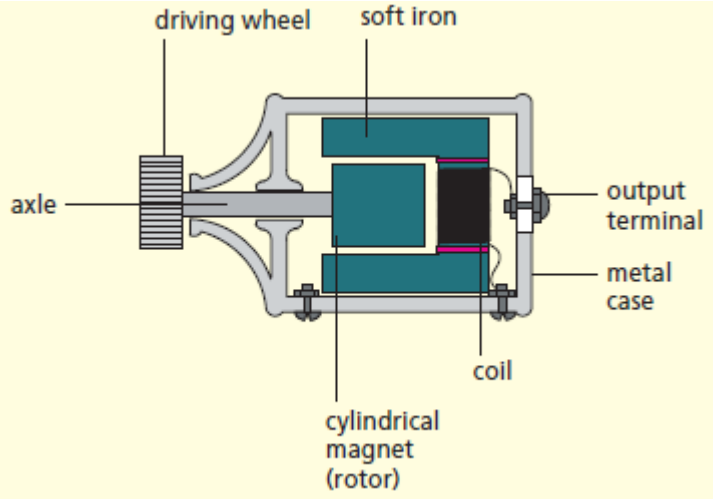
a-What are A and B called and what is their purpose?

b-What changes can be made to increase the p.d. generated?



5-Practical generators

A-Bicycles



The rotor of a bicycle generator is a permanent magnet and the voltage is induced in the coil, which is at rest.

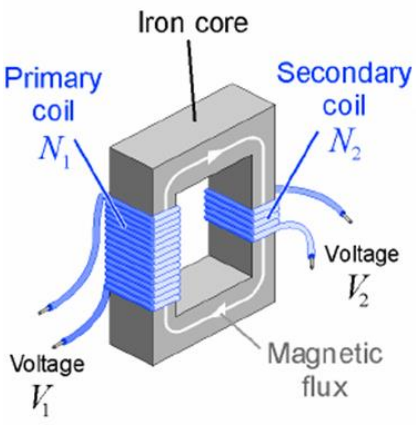
B-Power stations

In power station alternators the electromagnets rotate (the rotor) while the coils and their iron core are at rest (the stator). The large p.d.s and currents (e.g. 25 kV at several thousand amps) induced in the stator.

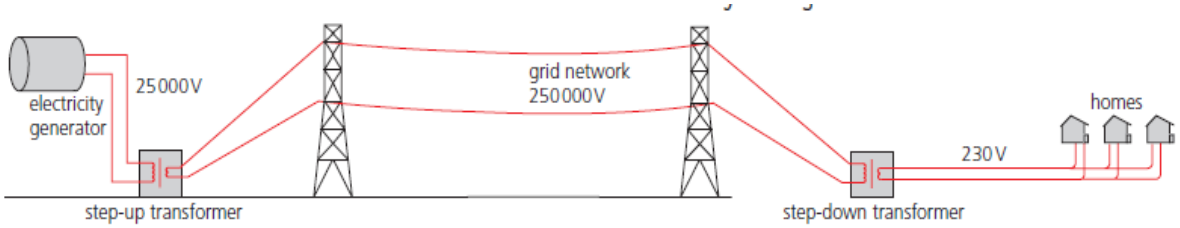


6-Transformers

Electrical Power Transformer is a static device which transforms electrical energy from one coil to another without any direct electrical connection to windings. It transforms power from one circuit to another without changing its frequency but may be in different voltage level. Transformers only work using an ac supply.



Ideal Transformer Model is developed by considering a transformer which does not have any loss.



☞A relationship between voltages and turns for a transformer results because the two coils have a different number of turns.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Example: Calculate the number of turns required for the primary coil of a transformer if secondary has 400 turns and the primary voltage is stepped up from 12V to a secondary voltage of 48V.

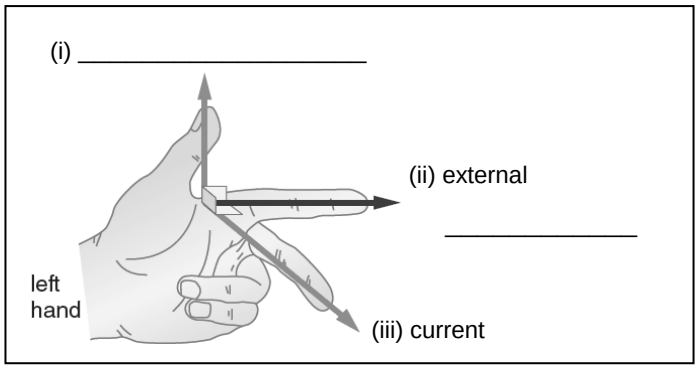
☞If a transformer is 100% efficient (ideal) then the power input to the primary coil is equalled by the power output from the secondary coil.

$$\frac{V_s}{V_p} = \frac{I_p}{I_s}$$

Example: Calculate the primary current if when a transformer is supplied with 230V the secondary provides 4A at a voltage of 13V. Assume that the transformer is 100% efficient.

Questions-Exercises

1- A current-carrying conductor experiences a force when placed in a magnetic field. The directions of the force, the magnetic field and the current can be worked out using Fleming's left-hand rule.



Hold the thumb and the first two fingers of the left hand at right angles. The thumb gives the direction of the magnetic force F, if the first finger points in the same direction as the external magnetic field B, and the second finger in the same direction as the current I.

2- Electric motor is a device which converts _____ energy to _____ energy.

3- The turning effect on a current-carrying coil inside a magnetic field can be increased by:

i. _____ the current,
ii. _____ the number of turns in the coil,
iii. _____ the area of the coil (within B-field),
iv._____ the strength of B-field.

4-A simple d.c. motor (Fig 4w) contains a rectangular coil which is free to rotate between the poles of a magnet. Current flows into the coil via a pair of _____ and a _____(or split ring).

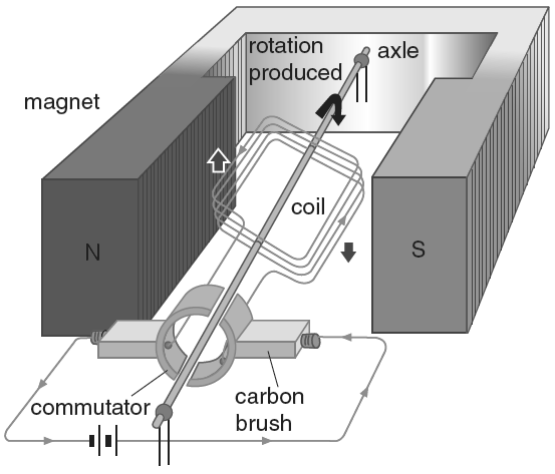
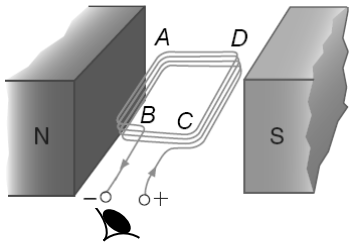


Fig 4w

5-A rectangular current-carrying coil between two magnetic poles is shown in Figure a.

(a) State the direction of the magnetic force acting on each side of the coil.
(b) What is the direction of rotation of the coil as seen by the observer?
(c) Why does the coil finally stop and lie along the vertical?
(d) How can the above set-up be turned to a d.c. motor?
(e) Explain how the changes in (d) making the coil rotate continuously?



Side *Fig a*

Answer:

(a)- Side AB:_____

Side BC:_____ CD:_____

Side AD:_____

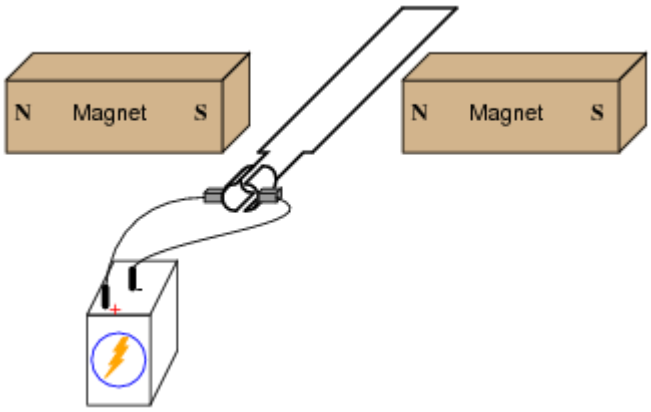
(b)- _____

(c)- When the coil is vertical, the magnetic forces act along the same line and _____ _____ each other, and the turning effect becomes _____. Therefore, the coil finally stops and lie along the vertical.

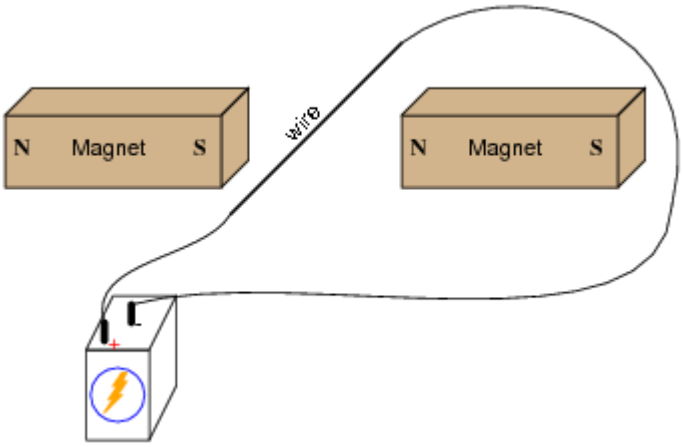
(d)- Connect the two ends of the coil to the power supply via _____ and _____.

(e)- The commutator reverses the _____ through the coil when it changes contact from one brush to the other. This also reverses the _____ acting on the coil. Therefore, the coil keeps rotating in the same direction.

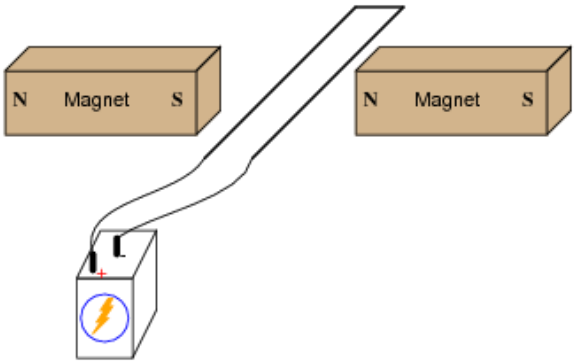
6-If the ends of a wire loop are attached to two half-circular metal strips, arranged so that the two strips almost form a complete circle, and those strips are contacted by two "brushes" which connect to opposite poles of a battery, which way will the wire loop rotate?



7-If an electric current is passed through this wire, which direction will the wire be pushed (by the interaction of the magnetic fields)?



8-If current is passed through a loop of wire, as shown, which direction will the loop rotate?



9- A transformer has a primary coil of 2000 turns and a secondary of 400turns.

(a) Is this a step-down or step-up transformer?

(b) If the primary coil is supplied with a voltage of 500V what is the secondary voltage?

(c) If the secondary coil has a voltage of 12V what is the primary voltage?

10. A transformer has a primary coil of 6000 turns and a secondary of 150 turns.

(a) Is this a step-down or step-up transformer?

(b) If the primary coil is supplied with a voltage of 230V what is the secondary voltage?

(c) If the secondary coil has a voltage of 9V what is the primary voltage?

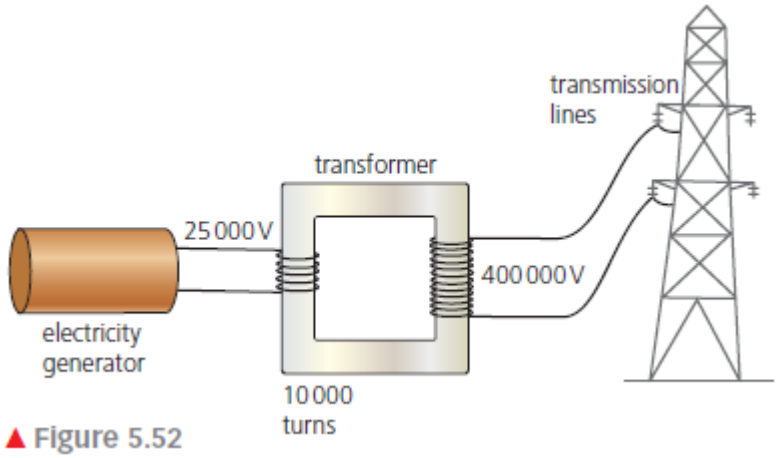
11. A transformer has a primary coil of 400 turns and a secondary of 1600turns.

(a) Is this a step-down or step-up transformer?

(b) If the primary coil is supplied with a voltage of 30V what is the secondary voltage?

(c) If the secondary coil has a voltage of 800V what is the primary voltage?

12. The generator in a power station produces electricity at a potential difference of 25 000 V. A transformer steps this up to 400 000 V, for the electricity to be transmitted on the national grid. See Figure 5.52 below.



▲ Figure 5.52

a) Use the information in the diagram to calculate the number of turns in the secondary coil of the transformer.

b) The current in the primary coil is 800 A. Calculate the current in the transmission lines.

c) Why are high potential differences used to transmit current on the national grid?